From Nutrition Goals to a Grocery Trip

This tool turns a health goal into a practical grocery list, a rough cost estimate, and nearby store suggestions.

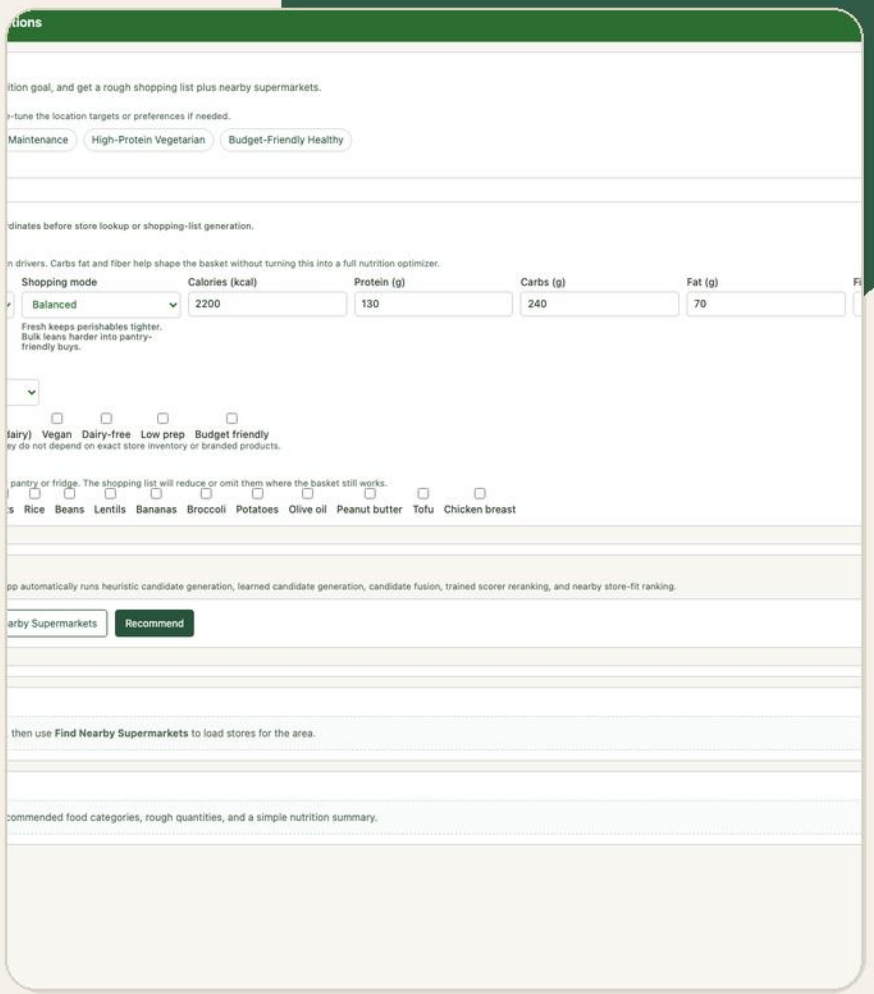
Non-technical audience first

Observed on the deployed app

Focus: decisions, tradeoffs, outcomes

What the audience sees

- Preset goals like muscle gain, fat loss, maintenance, and budget-friendly healthy.
- A one-click path from location and targets to a shopping list and store suggestions.
- Outputs that read like a grocery assistant, not a technical dashboard.



Hero view: the deployed dashboard starts with presets, local store lookup, and

Who This Is For

The strongest non-technical audience is not a data scientist. It is a shopper who wants a healthier and easier grocery trip.

Primary user persona

Busy students and young professio

People who want to eat for a goal such as muscle gain, fat loss, maintenance, or a healthier budget without hand-building a meal plan.

Fast setup

They want a recommendation in minutes, not a spreadsheet exercise.

Affordable

They care about rough basket cost before they get to the store.

Local

They need recommendations that feel tied to nearby stores.

Understandable

They need outputs they can trust without knowing the math.

Why that audience fit makes sense

The app exposes presets, shopping modes, pantry checkboxes, nearby store lookup, and plain-language notes. That is the design of a guided consumer decision tool, not a specialist analytics console.

The Real-World Problem

Nutrition advice is abstract, but shopping is local, budget-constrained, and full of practical tradeoffs.

Abstract advice

A user may know they want more protein or a lower-cost basket, but that does not tell them what to buy.

Too many tradeoffs

Calories, protein, produce, convenience, pantry items, and cost all pull the basket in different directions.

Local reality

The grocery decision depends on nearby stores and regional price levels, not just the target on paper.

Decision fatigue

Without structure, the user either guesses or buys a basket that does not actually match the goal.

Communication goal: help a non-expert user turn a health goal into a grocery decision they can actually act on.

What The Tool Does

The best non-technical explanation is a simple decision pipeline, not a math lecture.

1

1. User inputs

Location, targets, preferences, pantry, and shopping window.

2

2. Basket options

The planner builds several plausible grocery baskets.

3

3. Final choice

It picks the basket that best balances nutrition fit, practicality, and price-awareness.

4

4. Decision support

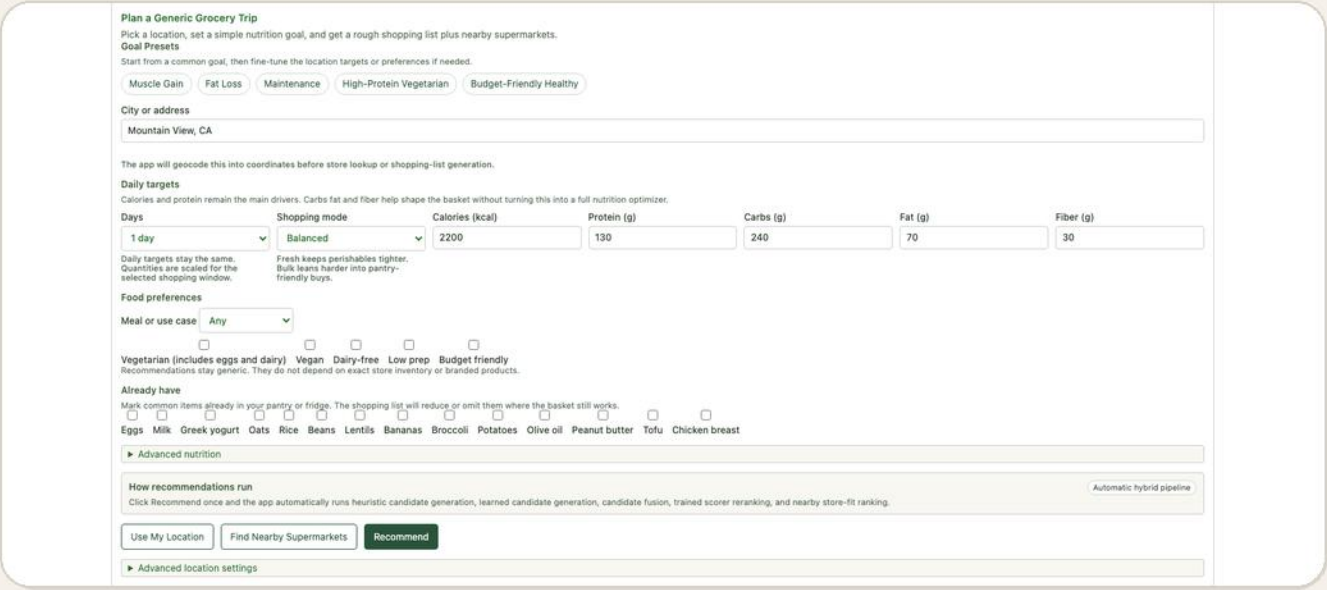
It returns a list, a summary, a rough cost, and store suggestions.

Plain-English summary

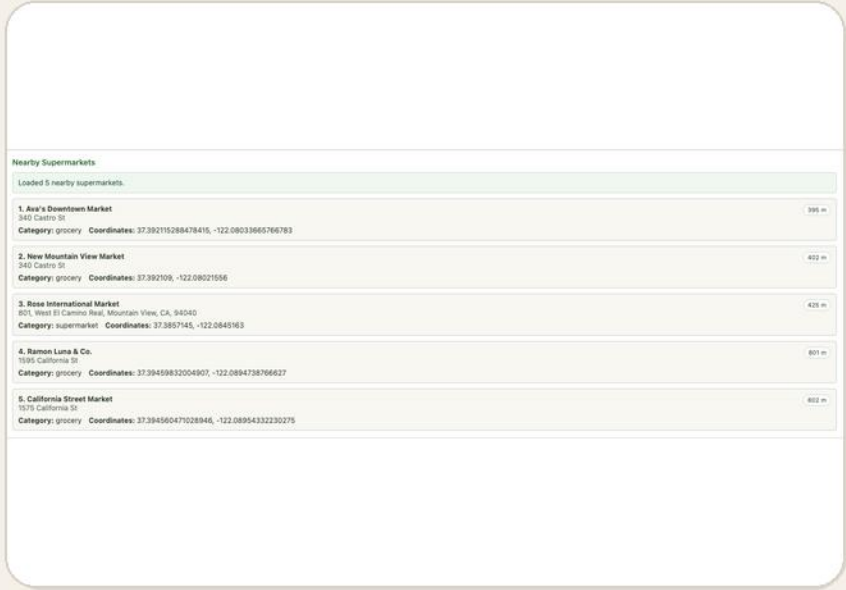
In other words, the app tries a few sensible shopping baskets and picks one that seems closest to the user's goal without ignoring realism.

How A User Interacts With It

The interface is organized like a guided grocery-planning workflow rather than a technical analytics dashboard.



Step 1: set the goal and preferences



Step 2: load nearby supermarkets

Why this is user-friendly

- Preset buttons reduce setup effort for common goals like muscle gain, fat loss, and budget-friendly healthy.
- The main controls are everyday choices: calories, protein, days, shopping mode, dietary flags, and pantry items.
- Nearby store lookup and Recommend are separate steps, which helps a first-time user explore before committing.

What Changes When Decisions Change

These outputs were observed on the deployed app in Mountain View while holding the same base target constant.

Shared setup: Mountain View, CA | 2,200 kcal | 130 g protein | 1 day | balanced shopping mode

Base plan

$8.54

Typical basket cost

Chicken breast | Rotisserie chicken | Wholemeal bread | Spinach | Carrots

Observed effect

Balances the target with familiar staples and a produce stop.

Protein gap: +7.7 g

Produce fit: La Plaza

Budget preference

$5.44

Typical basket cost

Lentils | Eggs | Rice | Cabbage | Carrots

Observed effect

Shifts toward cheaper pantry staples and keeps shopping simpler.

Protein gap: -19.4 g

Produce fit: Safeway

Low-prep preference

$9.39

Typical basket cost

Rotisserie chicken | Chicken breast | Wholemeal bread | Apples | Bell peppers

Observed effect

Leans into convenience foods while still staying close to the target.

Protein gap: +1.9 g

Produce fit: La Plaza

Takeaway: the app is not just filling calories and protein. It visibly trades off cost, convenience, and basket composition when the user changes preferences.

Key Insight / Recommendation Example

A concrete recommendation helps the audience understand what the tool is actually saying.

Rotisserie chicken is doing most of the protein work in this basket.
Swap option: Chicken breast Similar protein anchor that fits the same basket role.
Protein: 40.5 g **Calories:** 298.2 kcal
Typical regional price: $0.7 per 100 g; typical item cost about **$0.98** (regional range $0.72-$1.1).

Carb base 1 item

3. Wholemeal bread Carb base
Suggested buy: ~420 g Wholemeal bread
Easy carb base
Provides about 44 g carbohydrate per 100 g. Helps push the basket toward the fiber target.
Value note: Affordable fiber source Among the priced carb options, this one adds useful fiber without pushing cost up much.
Wholemeal bread provides an easy calorie and carbohydrate base.
Swap option: Oats Can replace the main carb base while keeping the basket balanced.
Protein: 35.2 g **Calories:** 1012.2 kcal
Typical regional price: $0.78 per 100 g; typical item cost about **$3.28** (regional range $2.35-$3.49).

Produce 2 items

4. Spinach Produce
Suggested buy: 1 bag Spinach (~260 g)
Practical produce pick
Adds produce with strong vitamin C coverage.
Value note: Good vitamin C value This produce choice gives strong vitamin C coverage for the average-price reference.
Spinach keeps the basket practical with a simple produce option.
Swap option: Bell peppers Easy produce swap with a similar support role.
Protein: 6.8 g **Calories:** 74.6 kcal
Typical regional price: $0.61 per 100 g; typical item cost about **$1.57** (regional range $1.29-$2.67).

5. Carrots Produce
Suggested buy: ~3 servings Carrots (~180 g)
Practical produce pick
Keeps the basket practical with an everyday produce option.
Value note: Affordable fiber source This produce item adds fiber at a relatively good average-price value.
Carrots keeps the basket practical with a simple produce option.
Swap option: Bell peppers Easy produce swap with a similar support role.
Protein: 1.1 g **Calories:** 72.4 kcal
Typical regional price: $0.38 per 100 g; typical item cost about **$0.69** (regional range $0.58-$0.73).

Extras / boosters 1 item

6. Olive oil Calorie booster
Suggested buy: ~3.6 servings Olive oil (~50 g)

Observed base recommendation

What the model is saying

Buy a small set of versatile staples that gets you close to the target without making the trip expensive or overly complicated.

Protein	137.7 g vs 130.0 g target
Calories	2160.4 vs 2200.0 target
Cost	About $8.54 for the basket

Why this matters

The recommendation is useful because it is actionable: it tells the user what to buy, what the basket roughly achieves, and where it may fit best nearby. In this observed case, Safeway is the one-stop pick and La Plaza is the stronger produce fit.

Limitations + How To Improve

Strong communication includes being honest about uncertainty, assumptions, and what the current tool still cannot do.

Current limitations

- The basket uses generic foods, not exact branded products or exact live store inventory.
- Price estimates are regional references from public price tables, not store quotes.
- Store recommendations are coarse fit suggestions based on nearby store type, brand, and distance.
- Long shopping windows are heuristic. Fresh and bulk modes both soften perishables, so results become approximate.
- The tool is not medical nutrition advice and does not capture allergies or complex health needs.

How to improve it

- Add real store inventory and current store-specific prices where possible.
- Let the user set a hard spending cap, not just a budget-friendly preference.
- Support allergies, household size, cultural preferences, and stronger nutrition guardrails.
- Collect feedback from real student or young professional users and adapt the interface around what confuses them.
- Show confidence labels more clearly so users know what is exact versus approximate.

Observed evidence: in 7-day mode, fresh landed at 631.6 g protein versus a 910.0 g target, and bulk landed at 780.2 g versus 910.0 g. That shows why perishables and scaling remain a real limitation.

Stakeholders + Ethics

A useful recommendation system should help without overstating certainty or excluding the people who need help most.

Shoppers

They gain a clearer starting point for cost, nutrition, and store choice, especially if they feel overwhelmed by grocery planning.

Stores and educators

Nearby stores may benefit from more intentional trips, and nutrition coaches or campus wellness groups could use the tool as a teaching aid.

Potentially underserved users

People with allergies, medical conditions, low food access, or strong cultural constraints may need more safeguards than the current tool provides.

Ethical design responsibility

The tool should keep telling the truth about approximation: regional prices are not store quotes and store-fit suggestions are not inventory guarantees.

Thoughtful ethical stance

The responsible position is to present the tool as a practical starting point for grocery decisions, not as medical advice or a perfect answer. Transparency is part of the product, not a disclaimer added at the end.

Takeaways + Future Use

The strongest close is simple: this project makes nutrition goals more actionable for an everyday shopper.

Why it matters

It turns an abstract nutrition goal into groceries, prices, and store choices a person can immediately use.

Why it is worth using

The outputs are understandable, local-feeling, and honest about what is approximate.

What comes next

Pilot it with real users, collect feedback, and strengthen the data and confidence signals.

Closing line

If the goal is better grocery decisions for non-experts, this tool already works as a strong communication bridge between nutrition targets and real shopping behavior.

Demo Backup

If the live demo fails, this backup slide still lets the audience see the full recommendation workflow and outputs.

What I would point to

- The grouped shopping list by food role.
- The cost estimate and price notes.
- The recommended store picks for the basket.
- The nutrition summary versus target.
- The approximation notes that prevent overconfidence.